

Sheilla Mumbi Macharia

Data Scientist · Applied Mathematics

PROFESSIONAL SUMMARY

Data Science practitioner with a background in Applied Mathematics and hands-on experience across predictive modeling, backend automation, and AI-powered reporting. Skilled in Python, FastAPI, and Streamlit for deriving actionable insights and shipping data-driven applications — from notebook-based analysis to scheduled pipelines and production APIs. Passionate about turning raw data into meaningful stories that drive smarter decisions, with proven success in projects spanning health prediction, agency workflow automation, retail analytics, and aviation risk assessment.

TECHNICAL SKILLS

Languages & Core: Python, SQL, HTML/CSS, Git, GitHub

Data Science & ML: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook

Backend & Automation: FastAPI, Flask, PostgreSQL, Jinja2, Gemini API

Visualization & BI: Tableau, Power BI, Streamlit, Plotly, Excel

Frontend: React, Tailwind CSS

Core Skills: Data Cleaning, Feature Engineering, Predictive Modelling, KPI Design, Sentiment & Time-Series Analysis, Business Intelligence

PROJECTS

AutoBrief — Automated Multi-Department Reporting Engine

Repository [↗](#)

Python · Flask · Gemini API · Jinja2

- Built a config-driven reporting engine that pulls data, computes KPIs, and generates narrated insights via the Gemini API across six departments (Sales, Finance, Marketing, Support, HR, Supply Chain) off one shared pipeline.
- Designed the AI-insight step to narrate pre-computed KPIs only, never asking the model to do arithmetic, keeping numbers reliable.
- Shipped both a CLI tool and a Flask + HTML/CSS/JS dashboard for department-level reporting.

Tafsiri — AI Client Workflow Automation for Creative Agencies

Repository [↗](#)

Python · FastAPI · PostgreSQL · Gemini API — 5-person team project

- Built the Client Payment Risk / Cash Flow Predictability engine as a FastAPI service within a 5-person team building an end-to-end AI agency-workflow tool (discovery calls, proposal generation, quoting, manager approval, and QuickBooks integration).
- Identified and resolved two data leakage issues that were silently inflating model performance.
- Debugged backend infrastructure issues (import-order bugs, Postgres sequence desync) and documented fixes via detailed PRs.

Kenya Regional Crop Yield Prediction

Repository 

Python · Scikit-learn · Pandas · NumPy · Matplotlib

- Developed an agricultural forecasting pipeline to predict regional crop yields in Kenya by integrating climate data, pesticide usage, and historical production metrics.
- Consolidated multi-source data from Kaggle, HarvestStat, and OpenAfrica, resolving spatial mismatches and engineering custom seasonal climate features (Long vs. Short Rains).
- Implemented and optimized Random Forest and XGBoost regressors, outperforming the baseline Linear Regression model by capturing complex, non-linear interactions between rainfall and crop output.
- Deployed the final model as a web application via Flask on Render, providing actionable forecasts to support early-warning food security planning and resource allocation.

SyriaTel Customer Churn Prediction

Repository 

Python · Scikit-learn · Pandas · NumPy · Matplotlib · Seaborn

- Built an end-to-end machine learning classification pipeline to predict customer attrition and safeguard telecom revenue using historical account data.
- Engineered data pipelines to drop non-actionable columns, encode binary plan metrics, and apply stratified splitting to manage class distributions cleanly across 3,333 records.
- Evaluated baseline models against Logistic Regression and a fine-tuned Decision Tree Classifier, choosing the tree architecture to achieve the best balance of interpretability and performance.
- Optimized the model to maximize Churn Recall (64%) and Precision (76%) at a 92% overall accuracy rate, successfully filtering high-risk accounts to help retention teams proactively minimize revenue loss.

Supermarket Performance Optimization Pipeline

Repository 

Python (Pandas, NumPy, Statsmodels) · SQL · Tableau · Power BI

- Engineered an end-to-end retail data science framework to analyze the operational variables driving footprint efficiency and revenue velocity across store locations.
- Conducted bivariate OLS regression analysis to discover a deterministic linear relationship ($p=0.999$) between store area and inventory capacity, alongside identifying a traffic paradox where gross footfall decoupled from total sales volumes.
- Resolved traditional grouping errors by replacing rigid, hardcoded spatial boundaries with dynamic, dataset-driven percentiles, ensuring perfectly balanced classification metrics across varying store sizes.
- Built interactive dashboards in Tableau and Power BI to translate complex retail metrics into actionable stakeholder insights.

EDUCATION

Bachelor of Science in Applied Mathematics

Expected 2026

University of Nairobi

Data Science Bootcamp

Aug 2025 – Feb 2026

Moringa School

Intensive hands-on program in Python, machine learning, and data visualization.

Capstone projects: SyriaTel Customer Churn Prediction, Kenya Regional Crop Yield Prediction.

CERTIFICATIONS

Data Science Program

eMobilis Technological Institute

Python for Data Analysis

FreeCodeCamp

Machine Learning & Data Visualization

Kaggle

Languages: English (Fluent) · Swahili (Fluent)